

# ECE 402: Communications Engineering

## Homework 1 (Fall 2026)

Due: Thursday 9/3

1. *Fourier Series:* Find the Fourier series in complex exponential form of the signal

$$x(t) = \cos(3\pi t) + \sin(6\pi t) + 2$$

2. *Dirac Delta:* Evaluate

$$\int_0^\infty [1 + 2t + t^2] [\delta(t - 2) + \delta(t + 2)] dt$$

3. *Fourier Transforms:* Use Poisson's Sum Formula to find the Fourier *Transform* of

$$x(t) = \sum_{n=-\infty}^{\infty} \text{sinc}\left(\frac{t-n}{2}\right).$$

Now use  $X(f)$  to find a simpler expression for  $x(t)$  in the time domain.

4. *Power and Energy Signals:* Classify each signal as a power signal, an energy signal, or neither. For a power signal, find the normalized power; for an energy signal, find the normalized energy.

(a)  $\Pi(t)$

(b)  $\cos^2(2\pi t)$

(c)  $\Pi(t) \cos^2(2\pi t)$

5. **Project 1, Chapter 1:** Consider again the signal from Problem 1:

$$x(t) = \cos(3\pi t) + \sin(6\pi t) + 2.$$

Use MATLAB to plot this signal from  $-1$  to  $+1$ , along with the Fourier Series you found in Problem 1. Be sure to label your axes and add a legend to distinguish the plots. Turn in your code and the plot.